

In-depth Explanation of Green's Functions: Application to a Metal Slab

1 Fundamental Concept

Green's functions are a powerful mathematical tool used to solve inhomogeneous linear differential equations. The core idea is to find the system's response to a point source (the "impulse response") and then use superposition to find the response to an arbitrary source.

Given a linear differential operator $\mathcal{L}$, we wish to solve:

$$\mathcal{L}u(x) = f(x) \quad (1)$$

The Green's function $G(x, s)$ is defined as the solution to:

$$\mathcal{L}G(x, s) = \delta(x - s) \quad (2)$$

where $\delta(x - s)$ is the Dirac delta function.

2 The Integral Solution

The general solution $u(x)$ is obtained by integrating the source $f(s)$ against the Green's function:

$$u(x) = \int G(x, s) f(s) ds \quad (3)$$

This effectively represents $\mathcal{L}^{-1}$ as an integral operator.

3 Case Study: The Metal Slab

Consider a metal slab of thickness L , occupying the region $0 \leq x \leq L$. We assume steady-state conditions (e.g., heat or electrostatics) where the potential or temperature $u(x)$ satisfies:

$$\frac{d^2u}{dx^2} = f(x) \quad (4)$$

3.1 Boundary Conditions

For a slab with fixed edges (Dirichlet conditions):

$$u(0) = 0, \quad u(L) = 0 \quad (5)$$

3.2 Piecewise Construction

Since the source is only at $x = s$, the solution is linear in the regions $x < s$ and $x > s$. Applying continuity at $x = s$ and the derivative jump condition:

$$G(x, s) = \begin{cases} \frac{x(s-L)}{L} & 0 \leq x \leq s \\ \frac{s(x-L)}{L} & s \leq x \leq L \end{cases} \quad (6)$$

3.3 Eigenfunction Expansion

Alternatively, we can express the Green's function using the orthonormal eigenfunctions $\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ and eigenvalues $\lambda_n = -\left(\frac{n\pi}{L}\right)^2$:

$$G(x, s) = - \sum_{n=1}^{\infty} \frac{2}{L} \frac{\sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi s}{L}\right)}{(n\pi/L)^2} \quad (7)$$

4 Summary of Properties

Property	Description
Boundary Conditions	$G(x, s)$ must satisfy homogeneous BCs of the original problem.
Reciprocity	$G(x, s) = G(s, x)$ (The effect of s on x equals the effect of x on s).
Singularity	The first derivative has a unit jump at $x = s$.
Application	Used in Electromagnetics, Quantum Propagators, and Heat Transfer.